

A Unified and Fully Automated Framework for Wavelet-Based Attacks on Random Delay

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Abstract—As a common defense against side-channel attacks, random delay insertion introduces noise into the executive flow of encryption, which increases attack complexity. Accordingly, various techniques are exploited to mitigate the defense effect of such insertions. As an advanced mathematical technique, wavelet analysis is considered to be a more effective technology according to its detailed and comprehensive interpretation of signals. In this paper, we propose a unified and fully automated wavelet-based attack framework (denoted as UWAF), whose data processing is kept within one unified wavelet domain, with three enhanced components: denoising, alignment and key extraction. We put forward a new idea of combining machine learning with wavelet analysis to realize the full automation of the program for attack framework, rendering it possible to search exhaustively for the optimal combination of parameter settings in wavelet transform. Our proposal finds a new setting of wavelet parameters that have not been exploited ever before and achieves the performance enhancement for about 20 times fewer traces required for successful key recovery. UWAF is compared with several mainstream attack frameworks. Experimental results show that it outperforms those counterparts, and can be considered as an effective framework-level solution to defeat the countermeasure of random delay insertion.

Index Terms—Random delay insertion, wavelet-based attack, machine learning, CPA.

I. INTRODUCTION

SIDE-CHANNEL Attack (SCA) utilizes the unavoidable but informative physical leakages during the encryption process to accomplish the key recovery. Such leakages may be time, power consumption, electromagnetic radiation, etc. The strategy *hiding* [1] is one of the main defense schemes against SCA. In the research line of *hiding*, the concept of *Random Delay Countermeasures* (RDC) was first proposed in [2]. And in 2020, [3] further divides the RDC into three types by specific implementation methods—*Random Propagation Delay* (RPD), *Random Delay Insertion* (RDI) [4] and *Random Clock Jitters* (RCJ). The first two are mainly implemented in software, and the last one is hardware-specific. In this paper, we are more interested in and concentrate on the software-based countermeasure RDI. And the attack effectiveness of our framework against RPD and RCJ will be discussed in Section VI.

A. Related Work

RDI often realizes the delays through dummy operations such as *NOP* instructions. In 2005, the first gate-level RDI against DPA was proposed in [4], which randomizes the instant power consumption and total charge quantity. Besides the hardware implementation, Tunstall et al. proposed an embedded software solution for RDI, decreasing the synchronization with less time as compared to those delays whose length follows a uniform distribution [5]. In CHES 2009, a new method of constructing and inserting random delays called Floating Mean (FM) was proposed in [6], which generates random delays with a much greater variance (yet with the same mean). Note that a greater variance indicates the increasing attack difficulty for adversaries, while the same mean implies that the computational cost remains unchanged. However, it was pointed out in [7] that the selected parameters of FM were inappropriate. Therefore, an enhanced scheme called Improved Floating Mean (IFM) was recommended, which is adopted in our experiments. Both FM and IFM are quite lightweight and can be easily realized at the software level, which are still extensively utilized in the SCA community.

With the continuous development of generation methods for random delay insertion (RDI), numerous attack strategies are proposed from different angles to break such countermeasure. According to the attack domain, current strategies against RDI can be divided into three types.

The first type is to attack RDI in the time domain. Such attacks tend to pre-align the traces so that the power

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consumption of target operations can be analyzed accumulatively, which facilitates DPA/CPA for key recovery. The first well-acknowledged alignment method—*static alignment* was proposed by Mangard et al. [1], which is easy to perform but difficult to work with complex delays. Afterward, Woudenberg et al. [8] proposed an enhanced method—*elastic alignment*, making up for the weakness of static alignment. Elastic alignment outperforms static alignment in aligning but is accompanied by a very large computational cost. In order to balance the effectiveness and the cost, Muijters et al. proposed the *rapid alignment* method in [9]. Additionally, Strobel et al. [10] attempted to identify the pattern of random delay by discriminating the Hamming Weight (HW) of the opcode derived from random delays and normal encryption operations. And Durvaux et al. [11] utilized the Hidden Markov Models to identify and eliminate the delays. Hence both of them require some prior knowledge about the target to model the delay-related (resp., encryption-related) instructions. To our knowledge, other time-domain attack methods like [12], [13], [14], [15], [16], [17], [18], [19] have been proposed in the past two decades as well. To date, more advanced techniques are applied in the SCA community, such as the deep learning technique. Particularly, neural networks are utilized to break RDI-protected implementations and most of them made an effort to construct end-to-end profiling attacks [20], [21], [22], [23], [24] with no requirement for pre-align operations.

The second type is to attack RDI in the frequency domain. The general idea is to convert the traces from time domain to frequency domain, and the key extraction is directly conducted in frequency domain. Recall that RDI inserts certain noise into the traces in time domain. Thereby, the resulting shifts will only affect the phase spectrum in frequency domain, instead of the amplitude spectrum. Based on the power spectral density, Gebotys et al. and Schimmel et al. proposed *Differential Power Frequency Analysis* and *Correlation Power Frequency Analysis* in [25] and [26], respectively. However, the performance of frequency-domain attacks is subject to the window size of *Discrete Fourier Transform* (DFT). In [27], it was proved that those attacks tend to fail when the maximum window size of DFT is smaller than the length of delays.

The third type is to attack RDI in the wavelet domain. As an advanced algorithm over Fourier transform, wavelet transform is becoming more and more popular in SCA research recently. In the SCA community, a very mature application of wavelet transform is wavelet-based denoising, which is exploited as a pre-processing mean for DPA/CPA. For example, Liu et al. and Ai et al. proposed their method to improve the signal-to-noise ratio in [28], [29], respectively. Besides, wavelet analysis was also utilized to denoise the misaligned signals [30]. Other than denoising, Debande et al. extended the idea and extracted the secret key directly from wavelet coefficients after wavelet transform in [31]. And [32] exploited wavelet scattering transform to align the traces. Furthermore, wavelet analysis has been combined with machine learning technology. For instance, in [33], wavelet transform was executed to pre-process power traces for *Probabilistic Neural Network* (PNN) training. And in [34], wavelet analysis and *Support Vector Machine* (SVM)

algorithm were combined to reveal the key in AES implementations. Interestingly, Zhang et al. [3] proposed a new method for wavelet-based alignment, which is combined with three components in the wavelet domain to construct the so-called *Wavelet-based Attack Framework* (WAF). It is verified to outperform the time-domain and frequency-domain counterparts (both for individual components and the whole framework). However, there exist some weaknesses in WAF and thus we intend to devise a new wavelet-based attack framework, which makes up for the shortcomings of WAF and gives full play of wavelet analysis against RDI.

B. Contribution

Our contribution can be summarized as below:

- **Unification:** We propose a *Unified Wavelet-based Attack Framework* (UWAF) against RDI, including three components: *Wavelet Trace Pre-processing* (WTP), *Wavelet Trace Alignment* (WTA) and *Wavelet Key Extraction* (WKE), which are all in the same wavelet domain. In Zhang's framework WAF [3], there exist two components that are in different wavelet domains. Hence, an extra back-and-forth transform is required between the time domain and wavelet domain, which incurs unnecessary errors accompanied by heavier computational overhead. The "unification" of our attack framework implies halved computational cost and more powerful attack ability.
- **Automation:** We improve the traces alignment component to achieve full automation for UWAF. The previous alignment in [3] exploits a threshold method, which requires manual interaction to ensure the initial range of the threshold and thereby hinders the automated flow of the attack program. To address it, we leverage the clustering algorithm of machine learning to synchronize the power traces. Under the premise of alignment accuracy, we have achieved full automation of the whole attack framework and mitigated the computational cost.
- **Exhaustion:** We recommend seeking out the optimal wavelet parameters for wavelet decomposition fully guided by the practical experimental results in absence of a sound theoretical model. And we highlight that the improved UWAF could perfectly fit the exhaustive search strategy. Hence, we perform an exemplary exhaustive search leveraging UWAF to find out the optimal wavelet parameters in our case. To diminish the computational cost brought about by exhaustion, we propose several methods to narrow the exhaustive scope, whose effectiveness is verified by experimental results in practice.
- **Evaluation:** To prove the improvement of UWAF and the validity of the exhaustion strategy, we evaluate the overall performance of UWAF through practical experiments and compare it with several framework-level attack methods. Our experimental results demonstrate that the wavelet parameters Db7 and level 9 found by exhaustion indeed perform the best in our experimental set-up, and also prove that UWAF outperforms the other framework-level counterparts when confronted with RDI.

C. Organization

This paper is organized as follows: Section II provides the background knowledge of wavelet analysis and introduces the previous WAF proposed in [3]. Section III elaborates on the newly designed UWAF to highlight its automation and unification. In Section IV, we explain the reason to search exhaustively for optimal wavelet parameters, and an exemplary exhaustion is provided. Finally, UWAF is evaluated and compared with other framework-level attack methods in Section V. Section VI highlights the difference between WAF and UWAF and concludes the paper.

II. BACKGROUND

A. Wavelet Analysis

Wavelet Transform (WT) can be represented by a time-domain signal $f(t)$ using a wavelet basis function ψ with two parameters: scaling s and shift u . The wavelet basis function is illustrated in Equation 1, where scaling s indicates the expansion and contraction of the wavelet and shift u depicts the displacement of the wavelet. Equation 2 shows the details of WT. It is integrated over the product of target signal $f(t)$ and wavelet basis function ψ . Hence, in the convolution process of $f(t)$ and ψ , both the frequency information and the corresponding time-domain shift can be sustained according to appropriate s and u .

$$\psi_{s,u}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-u}{s}\right) \quad (1)$$

$$\text{WT}(s, u) = \frac{1}{\sqrt{s}} \int_{-\infty}^{+\infty} f(t) * \psi\left(\frac{t-u}{s}\right) dt \quad (2)$$

There are two types of wavelet transforms: Continuous Wavelet Transform (CWT) and Discrete Wavelet Transform (DWT). Since the power consumption is discrete samples collected by a digital oscilloscope in the SCA community, DWT is utilized as the main technique for trace processing. DWT can be also called wavelet decomposition. As demonstrated in Equation 1 and 2, the information contained in the signals after wavelet decomposition varies with different wavelet basis functions. Hence, the selection of wavelet basis function in DWT plays an important role when considering the performance of signal processing.

Wavelet basis functions can be classified into different families, such as Daubechies (DbN), Mexican Hat (Mexh), Morlet (Morl), Meyer (Meyr), Symmlets (SymN) and Coiflets (CoifN). The letter “N” in DbN, SymN and CoifN indicates the order of the wavelet basis function. For example, Db1 has the order of one. Two key factors should be considered when selecting wavelet basis functions. One is the property of the wavelet families, and the other is the similarity between the original signal and the reconstructed one [3].

As illustrated in Fig. 1, the wavelet decomposition of a signal is composed of two components: filtering and down-sampling. Filtering is conducted by both a Low Pass Filter (LPF) and a High Pass Filter (HPF). Hence, signals are accordingly divided into low-frequency signals and high-frequency ones. After filtering, those filtered signals should be down-sampled by two

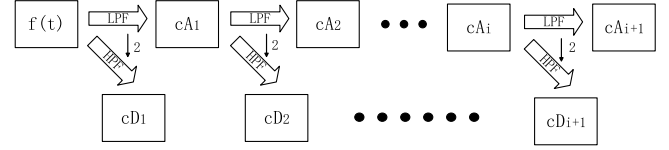


Fig. 1. Discrete wavelet transform (DWT) at multiple levels.

to remove redundant information. The process composed of both filtering and down-sampling is called one level of wavelet decomposition. However, the low-frequency signals can be continued to perform the next level of wavelet decomposition. Therefore, signal $f(t)$ will be continuously decomposed into two parts in the wavelet domain: the low-frequency ones called *approximation wavelet coefficients* (cA_i) and the high-frequency ones called *detail wavelet coefficients* (cD_i), where i implies the decomposition level. For example, if a three-level decomposition is applied to a signal, cD_1, cD_2, cD_3, cA_3 will be eventually derived. Note that cA_i generally represents the patterns and pivotal information of the signal, while cD_i usually contains noise and some irrelevant information. We highlight that the filtering operation can be regarded as the process of expanding the original signal with the selected wavelet basis function. Therefore, to perform an appropriate wavelet decomposition for better signal processing, careful selection for both the suited wavelet basis function and the specific decomposition level is required. In the rest of the paper, we refer to wavelet basis function and decomposition level as wavelet parameters.

B. Wavelet-Based Attack Framework

Zhang et al. [3] proposed a Wavelet-based Attack Framework (WAF) against Random Delay Countermeasures (RDC), including three main components: *Wavelet Trace Denoising* (WTD), *Wavelet Trace Alignment* (WTA) and *Wavelet Key Extraction* (WKE). The top of Fig. 2 depicts the high-level attack process of WAF. For the sake of description, we assume very ideally the physical leakage can be simply decomposed into three parts: normal encryption operations, noise, and random delays, though it is far more complicated in real circumstances. And we also show for brevity in the figure that the random delays part is superimposed on the normal encryption signals, which differs from the truth that the random delays are actually inserted in the encryption operations, causing a shift in time. Here we will give a brief introduction to WAF (for more details please refer to [3]).

As shown at the top of Fig. 2, the traces will first be converted from the time domain to the wavelet domain by DWT (marked as ①) for denoising. In the wavelet domain, WTD (marked as ②) removes the noise and the irrelevant information hidden in the cD_i by the classical Donoho formula [35]. After WTD, the updated values of cA_i and cD_i are utilized to re-construct the time-domain signals by IDWT (marked as ③). And then, the time-domain signals should be transformed into the wavelet domain again for alignment. During this DWT (marked as ④), the wavelet basis function and decomposition level are carefully selected to distinguish the normal encryption part and the

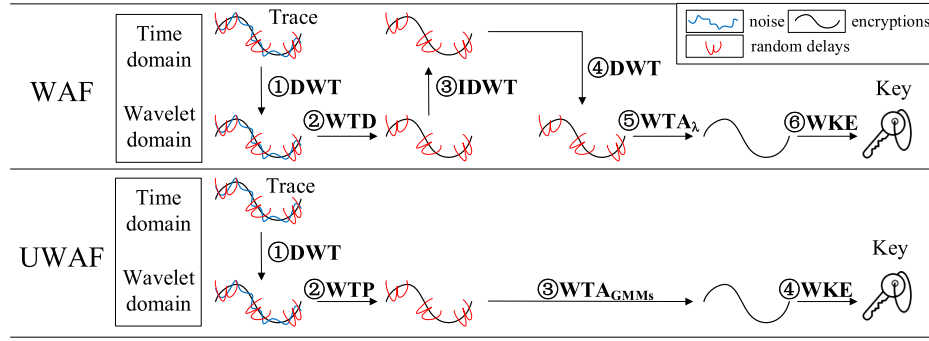


Fig. 2. High-level process of WAF and UWAF.

random delays part. Thanks to the proper selection of wavelet parameters, the random delays can be removed by the threshold method (denoted as WTA_λ , marked as ⑤) and the denoised traces could be finally well aligned. At this step, there is no need to switch to the time domain. CPA could be directly applied in the wavelet domain for key recovery, denoted as WKE (marked as ⑥).

With aforementioned components and steps, a complete attack of WAF is fully carried out. According to the systematic comparisons with counterparts in the time and frequency domain, WAF has been proved to perform better [3]. However, there still exist some weaknesses in this attack framework:

- The initial range of threshold in WTA_λ is required to be manually decided. After proper wavelet decomposition (marked as ④), the normal encryption part and random delays part could be distinguished in amplitude. That is, the maximum amplitude value of delays is smaller than the minimum amplitude value of encryption. Hence, the value range of the threshold is from the maximum amplitude of delays to the minimum amplitude of encryption. However, the authors of [3] give no details about how to determine those two boundary values. It seems that those values should be set manually by then, which blocks the automated execution flow of the attack program.
- There exists a redundant back-and-forth time-to-wavelet transform between WTD and WTA_λ , incurring an increase in computational cost and unnecessary errors. Recall that both WTD and WTA_λ require wavelet decompositions, and an extra inverse wavelet decomposition (marked as ③) is accordingly added between them. Since WTD and WTA_λ need different wavelet parameters, the computational cost of wavelet analysis is doubled. Moreover, the mandatory signal reconstruction is prone to bring about unnecessary errors.
- The reasons of wavelet parameter selection in wavelet decompositions (marked as ② and ④) are not that rigorous and convincing. In [3], the parameters for WTD are directly obtained by the conclusion from the previous literature work. While the parameters for WTA_λ are selected by theoretical derivations with no rigorous proof. Moreover, whether the different combinations of wavelet parameters could lead to an optimized solution under one framework is an open problem.

Therefore in this paper, we are strongly motivated to redesign the framework to address those questions. Our improvement can be summarized in three aspects. First, we fulfill the automated execution of the attack framework by improving the traces alignment component. Second, we unify the wavelet domain for all the components required to construct a complete attack framework. Finally, we recommend choosing the wavelet parameters fully guided by experimental results and give a practical paradigm.

III. UNIFIED WAVELET-BASED ATTACK FRAMEWORK

This section elaborates on the improved Unified Wavelet-based Attack Framework (UWAF). To begin with, a high-level overview of UWAF will be presented to show the general attack process. And then, the main components of UWAF will be detailed step by step.

A. Overview of UWAF

Recall that in WAF [3], two wavelet decompositions (DWT) with different wavelet parameters are required, which doubles the computational cost and introduces errors in the signal reconstruction process. Moreover, whether the various wavelet parameters for two DWT can jointly benefit the final key extraction is unknown. Therefore, we are strongly motivated to conduct all the wavelet-based analyses in the same wavelet domain, which we call “unify” and thus the proposed attack framework is accordingly termed as a “unified” framework.

Similar to WAF, it consists of three components, including *Wavelet Trace Pre-processing* (WTP), *Wavelet Trace Alignment* (WTA) and *Wavelet Key Extraction* (WKE), which are marked as ②, ③ and ④, respectively at the bottom of Fig. 2. It is obvious from Fig. 2 that the high-level difference between WAF and UWAF lies in the removal of once back-and-forth transform between the time domain and wavelet domain (IDWT and DWT).

As depicted at the bottom of Fig. 2 and Algorithm 1, UWAF will first convert the original signal (trace) $\mathcal{S}$ from the time domain to the wavelet domain by DWT (marked as ①). Here we assume very ideally for the sake of explanation (as claimed before) that $\mathcal{S}$ is originally composed of the encryption part $\mathcal{S}_{ep}(t)$, the random delay part $\mathcal{S}_{dp}(t)$ and the noise part $\mathcal{S}_{np}(t)$. After choosing the proper wavelet basis function f_{UWAF} and

Algorithm 1 Unified Wavelet-based Attack Framework

$$\begin{aligned}
& \mathcal{S} = \mathcal{S}_{ep}(t) + \mathcal{S}_{dp}(t) + \mathcal{S}_{np}(t) \\
& \text{DWT}(f_{\text{UWAF}}, l_{\text{UWAF}}) : \mathcal{S} \longrightarrow \mathcal{W} \\
& \mathcal{W} = \mathcal{W}_{ep}(w) + \mathcal{W}_{dp}(w) + \mathcal{W}_{np}(w) \\
& \text{WTP} : \mathcal{W} \longrightarrow \mathcal{W}^1 \\
& \mathcal{W}^1 = \mathcal{W}_{ep}^1(w) + \mathcal{W}_{dp}^1(w) \\
& \text{WTA}_{\text{GMMs}} : \mathcal{W}^1 \longrightarrow \mathcal{W}^* \\
& \mathcal{W}^* = \mathcal{W}_{ep}^*(w) \\
& \text{WKE} : \mathcal{W}^* \longrightarrow \mathcal{K}
\end{aligned}$$

decomposition level l_{UWAF} , original signal $\mathcal{S}$ is converted from the time domain to a new signal $\mathcal{W}$ in the wavelet domain. $\mathcal{W}$ also consists of the encryption part $\mathcal{W}_{ep}(w)$, the random delays part $\mathcal{W}_{dp}(w)$ and the noise part $\mathcal{W}_{np}(w)$ (ideally). Our goal is to eliminate the random delays $\mathcal{W}_{dp}(w)$ and the noise $\mathcal{W}_{np}(w)$ in the wavelet domain, so that the remaining encryption part $\mathcal{W}_{ep}(w)$ can be utilized directly for CPA¹. Hence, the component WTP (marked as ②) will be performed first to remove the noise part, resulting in that $\mathcal{W}$ turns to $\mathcal{W}^1$. Since the random delays part $\mathcal{W}_{dp}^1(w)$ remained in $\mathcal{W}^1$ could desynchronize the traces and reduce attack efficiency, WTA_{GMMs} ² (marked as ③) has to be conducted to eliminate $\mathcal{W}_{dp}^1(w)$ for alignment, resulting in the signal $\mathcal{W}^*$. Eventually, the trace $\mathcal{W}^*$ only contains the encryption part $\mathcal{W}_{ep}^*(w)$ and can be directly leveraged to reveal the secret key $\mathcal{K}$ in the wavelet domain. This key extraction process is denoted as WKE (marked as ④).

In the following sections, we will detail the three components (WTP, WTA and WKE). And the selection strategy of wavelet parameters (wavelet basis function f_{UWAF} and decomposition level l_{UWAF}) for DWT will be provided in Section IV.

B. Wavelet Trace Pre-Processing

After DWT, the time-domain traces will be converted to cA_i (approximation wavelet coefficients) and cD_i (details wavelet coefficients) at a specific level i . Recall that cA depicts the approximate shape of traces with low frequency and cD contains the details with high frequency. Although cA and cD both have secret information and irrelevant noise, it is commonly recognized that cA mainly contains the information of signals, while most of the noise is retained in cD . Hence WTP will discard all the detail wavelet coefficients cD and the rest of the attack framework will be conducted on cA_i only³.

This seems to be a slightly bold decision, which is though based on the following two reasons. The cost of discarding cD is the loss of information hidden in the cD . However, the information contained in the cD is very little and the attack experiment result of [36] proves that the attacks on cA and

$cA + cD$ present similar efficiency (Fig. 6 of [36]) when targeting at unprotected traces⁴. Hence, we stress that it is worth removing most of the noise at the cost of negligible loss of information if with a proper wavelet decomposition. Second, since we attempt to include WTP, WTA and WKE in the same wavelet domain, that is, the back conversion to the time domain is no longer required (IDWT of WAF) in the whole attack framework, there is no need to retain the cD for signal reconstruction. The effectiveness of the discarding strategy will be verified by our experimental results in Section V.

C. Wavelet Trace Alignment

In WAF [3], the component WTA exploits the threshold method to distinguish the normal encryption part and random delays part. Although the determination strategy of the threshold is effective, the initial value range of the threshold must be pre-set manually. This hampers the self-execution of the attack program and increases the attack cost (manual interaction and the resulting time cost), which is never desired. To address this issue, we apply a new approach to synchronize the traces in UWAF.

To compare the alignment performance fairly with the previous WTA of WAF, we follow the same attack flow and experimental setting as in [3]. That is, we will perform WTD of WAF first and then transform the denoised signals back to the time domain. Before performing the alignment approach, we will conduct the same DWT as in WAF. Note that those two wavelet decompositions leverage the same wavelet parameters with [3]. Hence, before alignment, the denoised time-domain signals should be transformed to wavelet domain by wavelet basis function Db9 and decomposition level 9.

Fig. 3(a) plots the cA_9 of one randomly selected trace⁵ after wavelet decomposition. It is demonstrated that the normal encryption part (marked by a blue box on the top) and the random delays part (marked by a red box at the bottom) can be differentiated after careful wavelet decomposition. Such characteristics of the parts distribution have also been reported in the original TIFS paper [3] under the same experimental settings. Intuitively, the alignment problem in such circumstance can be naturally solved by clustering technology of machine learning. Since our case is a simple binary classification, here we apply and compare two common and classical clustering algorithms: *K-means Clustering* and *Gaussian Mixture Models Clustering*. To avoid ambiguity, we use WTA_λ to denote the threshold method utilized in WAF, while $\text{WTA}_{\text{kmeans}}$ and WTA_{GMMs} represents the clustering by K-means and Gaussian Mixture Models, respectively.

1) *K-Means Clustering*: K-means clustering is an iterative clustering algorithm proposed in [37]. It can divide the data set into K clusters according to a specific distance function (Euclidean Distance in our case), while K is a required number as the input parameter. In our case, we set $K = 2$, since we should

¹This is also an ideal description. In practice, it is impossible to decompose signals (in time and wavelet domain) into the three mentioned parts perfectly.

²We use subscript “GMMs” to highlight that this component WTA_{GMMs} utilizes different approach from WTA_λ in WAF for alignment.

³Note that cD_i will be utilized to reduce the range of decomposition levels for the final exhaustive search (detailed in Section IV-C), which however is not involved in the attack framework.

⁴Since the traces will be synchronized before the direct CPA, the attack target can be regarded as unprotected in our experiments.

⁵This trace is obtained by the same experimental setup detailed in Section IV-A.

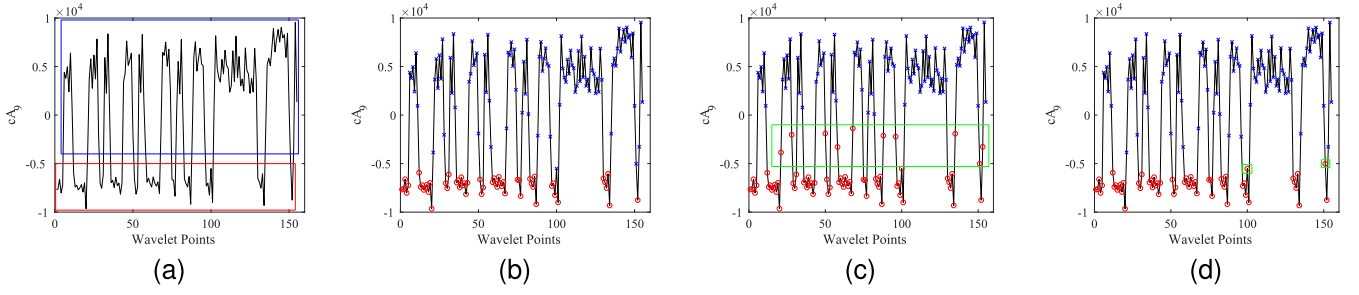


Fig. 3. Illustrations of cA_9 trace under different clustering methods.

identify the random delays part from the normal encryption part. Fig. 3(c) illustrates the clustering result of cA_9 (illustrated in Fig. 3(a)) by K-means method. In comparison, Fig. 3(b) plots the two groups clustered by WTA_λ for the same trace. For both figures, the points marked by the red circles correspond to the random delays, while those points marked by the blue crosses imply normal encryption operations. It can be observed by comparison that some points are misassigned into the random delays group in Fig. 3(c) (plotted by green box), however they actually correspond to the normal encryption part implied by Fig. 3(b). Hence, the alignment performance by leveraging K-means method is not as effective as the threshold method WTA_λ utilized in WAF. This is essential because K-means method achieves the clustering by distance, which infers that the shape of groups clustered by K-means method will be closer to a circle. The feature of the circle is that the two projections (on the x -axis and y -axis) share the same length. However, we can observe from Fig. 3(a) that the length of projections on the x -axis and y -axis for both the random delays group and encryption group are different, thereby prone to cause errors in K-means clustering process. And the middle points (plotted by the green box in Fig. 3(c)) are visually closer in distance to the center point of the random delay group, and thus be mistakenly marked as random delays, leading to poor alignment performance of K-means method.

2) *Gaussian Mixture Models Clustering*: Since the clustering principle of K-means method limits the alignment accuracy, we propose to utilize an advanced clustering method called Gaussian Mixture Models (GMMs) clustering. GMMs assumes that the data points belonging to one cluster follow a Gaussian distribution, which is featured by two parameters: mean μ and variance σ^2 . Hence, the clusters in any shape of ellipse can be accurately captured by GMMs. This characteristic exactly matches the points distribution in our case, where the two parts (normal encryption and random delays) are both ellipses with their major axis parallel to the x -axis. To find the mean μ and variance σ^2 of clusters, the *Expectation Maximization* (EM) optimization algorithm is utilized. The clustering result on the cA_9 (presented in Fig. 3(a)) by GMMs clustering method is illustrated in Fig. 3(d). In comparison with Fig. 3(b), it can be observed that the clusters obtained by GMMs clustering are nearly the same as those by threshold method WTA_λ of WAF. However, there are still a few points (marked by small green squares in Fig. 3(d)) assigned to different clusters by different

methods. With no prior knowledge, it is hard to determine which clustering result is more accurate. However, those controversial points are too few among the total amount of data points to cause a negative impact on the final alignment performance and thus can be omitted to some extent. Compared with Fig. 3(c), it is obvious that GMMs approach outperforms the K-means method since its Gaussian distribution simulation essentially better matches the points feature than the distance-based circles in our experiment.

Fig. 3 just illustrates one exemplary trace. However, our experimental results demonstrate that the observation holds for all the traces collected in our experiment. That is, the GMMs clustering method is able to improve the clustering accuracy in comparison with K-means method, meanwhile achieving a similar alignment effectiveness with the threshold method leveraged in [3]. Note that there are also other excellent clustering algorithms, which may be more proper for WTA, such as some approaches based on Non-parametric Estimation [38]. However, since GMMs already has the comparable ability with threshold method to identify delays for alignment, we do not keep delving deeper into the search for optimal clustering method in this paper and leave it as a future work. We should highlight that the advantage of GMMs clustering method over the threshold method WTA_λ is its automated execution when feeding with the desired number of clusters and the data points to be clustered. Therefore, taking GMMs clustering method in WTA, the new attack framework can work in an automated and effective fashion. The alignment stage of UWAF is fulfilled by discarding the points that are marked as random delays part after clustering.

We should underline that in the new attack framework UWAF, there is only one DWT before WTP. It implies that there is no extra DWT before WTA, and WTA is performed exactly after WTP in the same wavelet domain, which differs from WAF. The above experiments which conduct two wavelet decompositions (before WTD and WTA) are just to provide a fair and direct comparison of alignment performance.

D. Wavelet Key Extraction

After performing WTP and WTA in the same wavelet domain, the cA_i is well denoised and aligned, which is supposed to be vulnerable to side-channel attacks. Hence we can directly conduct Correlation Power Attack (with Hamming

Weight model) on the cA_i as in [3] and [36]. At this step, the new attack framework UWAF is fulfilled.

IV. EXHAUSTIVE SEARCH FOR WAVELET PARAMETERS

In this section, we focus on the selection strategy of wavelet parameters for wavelet decomposition. Compared with WAF, UWAF outperforms WAF in the automation and unification of the whole framework. “Automation” indicates that the attack flow is executed automatically with no manual interaction, while “unification” implies that all the wavelet-based analysis for the attack is conducted in the same wavelet domain (with the same wavelet parameters). Those features are conducive to the exhaustive search for wavelet parameters.

While exhaustive search for wavelet parameter selection might seem straightforward or unsophisticated at first glance, it is important to consider the following reasons that underscore its value and effectiveness. In the SCA community, the physical leakage varies and is affected by the encryption algorithm, implementation strategy, hardware platforms, acquisition devices, acquisition approaches, experimental environment, and some other unexpected factors. As a result, the wavelet-based analysis (i.e., the selection of wavelet basis function and decomposition level) on those observations should also consider the above-mentioned factors (such as sampling rate). Hence, it is an extremely tough task to construct a sound and practical theoretical model⁶ within a short period of investigation, since it should characterize the observations affected by complicated factors, and meanwhile combine with the wavelet transform, which is also a complex mathematical topic. Therefore, in the absence of leading theoretical analysis, it could be a good choice to be guided completely by experimental results, which at least can give solutions for the current problems. Furthermore, we should highlight that the optimal selection of wavelet parameters by exhaustive search is not only effective for the current attack, but also for the subsequent attacks in similar circumstances. For example, with the same hardware platforms and acquisition devices, targeting the same encryption implementations (which is a practical scenario), the selected parameters should remain efficacious (though perhaps not optimal owing to variations in environmental factors) for key recovery. Last but not least, from the evaluation perspective, it is also necessary to search exhaustively the possible wavelet parameters to find the worst attack scenarios, and give an impartial evaluation of the anti-SCA ability of the implementations.

According to the reasons explained above, we decide to search exhaustively the possible wavelet parameter combinations to find the optimal wavelet basis function and decomposition level in our experimental case. The weakness of exhaustive search is the expensive computational overhead, especially in wavelet-based analysis, since there are abundant wavelet families and deep decomposition levels (relying on the number of sampling points). To reduce the computational cost and render this method to be more practical, we will first show how to narrow the scope of exhaustion to a reasonable range. And

⁶It is certainly an ultimate goal for related research, which however requires a long and step-by-step investigation.

TABLE I
PROPERTIES OF SIX WAVELET FAMILIES

Property	Wavelet Family					
	Meyr	Mexh	Morl	DbN	CoifN	SymN
Compact support	no	no	no	yes	yes	yes
Orthogonality	yes	no	no	yes	yes	yes
Symmetry	yes	yes	yes	no	near	near
Vanishing moment	no	no	no	N	2N-1	N

then we will seek out the optimal wavelet parameters for our observations by exhaustive search. We believe that the ways to narrow the scope and the following exhaustive search experiment could be an example of wavelet-based attack against RDI-protected implementations leveraging UWAF. And the wavelet parameters we find could be effective (maybe not optimal) for key recovery in a similar experimental setup.

A. Experimental Set-Up

We choose AES-128 as the encryption algorithm since it is a common and classical symmetric block cipher. And the random delay insertions are generated as in [6] (Appendix B), [7] (Appendix B) and [3] (Fig. 2(b)), which is also available at Github [39]. Our attack target is the first byte of *SubBytes* of the first AES round. The AES-128 with RDI is implemented at the software level and runs on the ATmega163 micro-controller, deployed on the SASEBO-W [40]. To capture the power consumption of SASEBO-W, the Agilent DSO-X3034T oscilloscope is utilized with bandwidth at 350MHz and a maximum sampling rate of 5GSa/s (Samples per second). Note that the experimental set-up is the same as in [3] in order for a fair comparison of the attack performance. Totally, 9985 power traces are collected.

B. Coarse-Grained Screening of Wavelet Basis Functions

Recall that wavelet decomposition requires the selection of wavelet basis function and decomposition level. Hence we begin with the initial screening of wavelet basis functions.

As a beginning, we should filter the wavelet families according to specific properties. Six commonly utilized wavelet families are considered as candidates at the initial stage: Meyr, Mexh, Morl, DbN, CoifN and SymN. Table I lists the dominant properties of wavelet families, including compact support, orthogonality, symmetry and vanishing moment. Among them, compact support and orthogonality are basic characteristics that are required to be featured. However, symmetry which is often applied in image applications, is insignificant in our case. Last but not least, Vanishing Moment (VM) is an important metric, that illustrates the smoothness and concentration of a wavelet basis function in both the time and frequency domain [41]. The value of VM is preferred to be large since a larger value of VM indicates a smaller number of coefficients for inner product [41]. In Table I, the first three columns have the value of “no” in the first and last rows, so that wavelet families Meyr, Mexh and Morl are excluded from further consideration. To keep a large VM, DbN, CoifN and SymN remain for the second round of screening.

TABLE II
ED VALUES BY PERFORMING DBN WAVELET FUNCTIONS ON FIVE
RANDOM TRACES ($\times 10^{-7}$)

Db1	Db2	Db3	Db4	Db5	Db6	Db7	Db8	Db9	Db10
70.94	108.67	52.79	91.18	180.27	1928.83	214.52	5.52	4.83	3.61
69.32	105.89	51.85	88.88	176.49	1884.33	210.13	5.38	4.73	3.51
70.26	107.38	52.29	90.12	178.60	1907.74	212.57	5.46	4.78	3.57
70.03	107.45	52.19	90.10	178.33	1907.40	212.39	5.45	4.79	3.56
70.16	107.54	52.27	90.22	178.49	1909.58	212.70	5.45	4.80	3.56

TABLE III
ED VALUES BY PERFORMING SYMN WAVELET FUNCTIONS ON FIVE
RANDOM TRACES ($\times 10^{-8}$)

Sym1	Sym2	Sym3	Sym4	Sym5	Sym6	Sym7	Sym8	Sym9	Sym10
3683.14	306.49	105.92	490.74	429.86	60.91	3.01	7.40	5.73	24.22
3597.43	299.35	103.41	479.17	419.77	59.37	2.94	7.22	5.60	23.65
3637.08	302.85	104.73	484.71	424.85	60.01	2.98	7.31	5.66	23.95
3682.95	306.25	105.86	489.93	429.34	60.59	3.01	7.39	5.72	24.21
3650.75	303.29	104.79	485.34	425.28	60.00	2.98	7.31	5.67	23.97

TABLE IV
ED VALUES BY PERFORMING COIFN
WAVELET FUNCTIONS ON
FIVE RANDOM TRACES ($\times 10^{-6}$)

Coif1	Coif2	Coif3	Coif4	Coif5
5.53	64.49	1.38	205.82	47755.97
5.53	64.45	1.38	205.70	47709.77
5.59	65.11	1.40	207.91	48232.56
5.57	65.00	1.39	207.27	48073.38
5.46	63.73	1.37	203.18	47131.11

At this step, the wavelet basis functions belonging to those remaining wavelet families already have the properties that are desired for signal processing. Hence, the following selection of wavelet basis functions will be further enhanced by calculating the similarity between the original signal and the re-constructed one after wavelet decomposition [3]. This similarity indicates the ability of the wavelet basis function to save the information of the signal. Certainly, a larger similarity indicates more information captured in the wavelet domain, which stands a greater chance for key recovery. Such similarity is defined as Euclidean Distance (ED), shown in Equation 3:

$$ED = \sqrt{\sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3)$$

where n is the number of samples, while y_i and $\hat{y}_i$ are the samples of the original signal and the reconstructed signal, respectively. The smaller ED value implies higher similarity. Tables II~IV list the ED value of wavelet basis functions belonging to the DbN, SymN and CoifN, respectively. Note that the ED values are obtained by randomly selecting 5 traces (same for each wavelet basis function).

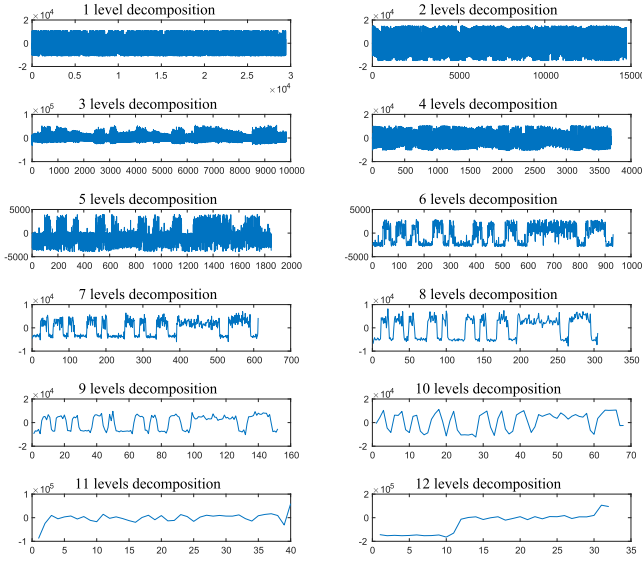
It is demonstrated from Tables II~IV that for each wavelet family, the order of N has a great impact on the signal similarity. In the DbN family, Db8~Db10 have a better similarity with the ED values all less than 10×10^{-7} . Similarly in the SymN family, Sym7~Sym10 present better similarity with their

ED value all less than 30×10^{-8} . Note that VM is proportional to the order N for those three families (DbN, CoifN and SymN), as presented in Table I. Therefore, both Db8~Db10 and Sym7~Sym10 with large N satisfy the two requirements simultaneously: larger VM and better signal similarity. However, according to the similarity, the better wavelet basis functions in the CoifN family are Coif1~Coif3 whose ED values are less than 100×10^{-6} . But considering the direct impact of the order N of CoifN on VM, for sufficiently large VM, Coif4~Coif5 are included for the final exhaustive search as well. Considering some fault tolerance, Db7 is also added to the candidates set due to its relatively large VM. Therefore, the candidates of wavelet basis function after this coarse-grained screening are Db7~Db10, Sym7~Sym10 and Coif1~Coif5.

C. Coarse-Grained Screening of Decomposition Levels

For DWT, once the wavelet basis function is selected, it requires determining the level which signal should be decomposed. Hence, the candidates of decomposition level should be reduced to a reasonable range as well. In our case, there are 12 levels to be filtered (denoted as L1~L12). The selection of number 12 relies on the amount of sample points. In theory, the levels can be added up until only one last point is left (recall that one level of decomposition means once down-sampling, which implies the number of sampling points halves). However in practice, the signal will be seriously distorted when decomposed to a certain level. Hence, the number 12 is an empirical selection that causes great distortion in our case (refer to the last picture of Fig. 4). This number can be easily obtained by performing one wavelet decomposition on an arbitrary trace.

To reduce the size of candidates from 12 levels, the *contrast Ratio of Approximation to Detail coefficients* (RAD) [3] is leveraged since it depicts the degree of divergence between the random delay parts and encryption parts. Note that in [3], RAD is utilized to directly determine the wavelet decomposition level for WTA_λ, while we use this metric to filter the decomposition levels. RAD is evaluated as in Equation 4, where $E(X)$ is the

Fig. 4. cA_i of one exemplary trace at different decomposition levels.TABLE V
RAD OF DIFFERENT WAVELET PARAMETERS COMBINATIONS

	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
Db7	1.00	0.93	0.89	0.90	0.74	1.12	2.23	2.22	2.03	1.62	1.27	0.94
Db8	0.99	0.93	0.91	0.90	0.77	1.13	2.20	2.17	2.02	1.60	1.26	0.90
Db9	0.98	0.93	0.91	0.92	0.73	1.13	2.16	2.12	1.95	1.58	1.29	0.86
Db10	0.97	0.93	0.92	0.94	0.78	1.15	2.12	2.07	1.87	1.50	1.19	0.80
Sym7	1.01	0.94	0.93	0.79	0.75	1.17	2.25	2.27	2.01	1.82	1.30	0.86
Sym8	1.02	1.03	0.94	0.77	0.72	1.15	2.33	2.31	2.05	1.74	1.25	0.91
Sym9	1.02	0.98	0.92	0.81	0.78	1.22	2.31	2.22	2.01	1.75	1.27	0.96
Sym10	1.01	1.03	0.94	0.77	0.73	1.20	2.25	2.20	1.92	1.64	1.22	0.93
Coif1	0.97	0.92	0.91	0.86	0.78	1.07	2.26	2.47	2.38	2.10	1.28	0.89
Coif2	1.02	0.95	0.90	0.79	0.78	1.09	2.42	2.51	2.28	2.02	1.31	0.86
Coif3	1.00	0.93	0.92	0.78	0.73	1.14	2.39	2.36	2.19	1.93	1.28	0.88
Coif4	0.97	0.95	0.92	0.80	0.76	1.17	2.34	2.27	2.07	1.83	1.27	0.89
Coif5	0.97	0.98	0.94	0.80	0.73	1.22	2.23	2.20	2.03	1.69	1.22	0.91

Expectation of X and $\max(X)$ (resp., $\min(X)$) returns the maximum (resp., minimum) value of X . Note that large RAD indicates that random delay parts and encryption parts are well separated, which facilitates the alignment operation. Table V records the RAD⁷ values for the wavelet parameters combined by the candidates of the wavelet basis function screened in Section IV-B and the 12 levels. In Table V, the rows correspond to the wavelet basis functions, while the columns are decomposition levels. It is indicated in Table V that L7~L10 can remain as candidates since they have larger RAD.

$$RAD = \frac{\sum | \frac{cA_i - E(cA_i)}{\max(cA_i) - \min(cA_i)} |}{\sum | \frac{cD_i - E(cD_i)}{\max(cD_i) - \min(cD_i)} |} \quad (4)$$

Now we will verify the effectiveness of RAD metric by practical attack experiments. Precisely, each wavelet parameters combination in Table V will be utilized to conduct a UWAF-based attack to test if it can succeed in revealing the key. If the first byte of the key can be successfully extracted, the attack can be marked as successful. In Table VI, the tick indicates that the associated wavelet parameters combination can extract the correct key byte, while the cross implies the attack fails.

⁷the RAD values in Table V differ from those in [3] (Fig. 6) since the denoising step differs between WAF and UWAF.

TABLE VI
RESULTS OF UWAF-BASED ATTACK WITH DIFFERENT WAVELET PARAMETERS COMBINATIONS

	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
Db7	×	×	×	×	×	×	✓	✓	✓	✓	×	×
Db8	×	×	×	×	×	✓	✓	✓	✓	✓	×	×
Db9	×	×	×	×	×	✓	✓	✓	✓	✓	×	×
Db10	×	×	×	×	×	✓	✓	✓	✓	✓	×	×
Sym7	×	×	×	×	×	×	✓	✓	✓	×	×	×
Sym8	×	×	×	×	×	×	✓	✓	✓	✓	×	×
Sym9	×	×	×	×	×	×	✓	✓	✓	✓	×	×
Sym10	×	×	×	×	×	✓	✓	✓	✓	✓	×	×
Coif1	×	×	×	×	×	×	✓	✓	✓	✓	×	×
Coif2	×	×	×	×	×	×	✓	✓	✓	✓	×	×
Coif3	×	×	×	×	×	✓	✓	✓	✓	✓	×	×
Coif4	×	×	×	×	×	✓	✓	✓	✓	✓	×	×
Coif5	×	×	×	×	×	✓	✓	✓	✓	✓	×	×

From Table VI, we can observe that almost all the wavelet basis functions combined with L7~L10 have the ability to recover the first key byte. Such attack results are consistent with RAD results in Table V, thereby providing strong evidence that RAD is able to perform an initial screening for the wavelet decomposition levels.

From Tables V and VI, we can also observe that the decomposition level plays a dominant role in the effectiveness of key recovery. In order to further explain this phenomenon, one trace is randomly selected and decomposed by Db7 at various levels (L1~L12) as an example. The decomposition results are listed in Fig. 4. The x -axis corresponds to those remaining wavelet points after down-sampling, while the y -axis represents the value of cA_i , $1 \leq i \leq 12$.

It is demonstrated in Fig. 4 that the random delays part and the normal encryption part can not be clearly distinguished at L1~L5. At L6, although the two parts can be differentiated to some extent, their distance is too close, which hinders the alignment performance. That is why there are more failed cases than successful ones at level 6 in Table VI. At L7~L9, the amplitude difference between the random delays part and the encryption part is quite significant, which benefits clustering and partition. And this is why key byte can be revealed with any wavelet basis functions among the candidates. At L10, the number of points for cA_i becomes insufficient, resulting in lower RAD values in Table V and one failed case in Table VI. At L11~L12, the over-compressed data causes serious distortion, which makes the associated attacks unsuccessful. In conclusion, the decomposition performance is consistent with our experimental results presented in both Tables V and VI as well. Therefore, according to both RAD results and decomposition performance, four levels (L7~L10) will remain after the coarse-grained screening.

D. Fine-Grained Screening of Wavelet Parameters

After the initial coarse-grained screening for both wavelet basis functions and decomposition levels, we intend to conduct a fine-grained screening to further ease the computational complexity of the final exhaustive search. To this end, we adjust the *Signal-to-Noise Ratio* (SNR) to illustrate the alignment performance of various wavelet parameter combinations. Our adjusted SNR is evaluated as in Equation 5, where “ $\text{Var}(X)$ ”

TABLE VII
SNR OF DIFFERENT WAVELET
PARAMETERS COMBINATIONS

	L7	L8	L9	L10
Db7	0.07	0.14	0.33	0.04
Db8	0.09	0.10	0.29	0.07
Db9	0.05	0.11	0.31	0.08
Db10	0.07	0.12	0.17	0.07
Sym7	0.06	0.06	0.04	0.03
Sym8	0.04	0.04	0.03	0.03
Sym9	0.04	0.06	0.11	0.03
Sym10	0.03	0.06	0.04	0.04
Coif1	0.05	0.05	0.04	0.03
Coif2	0.04	0.04	0.03	0.06
Coif3	0.07	0.11	0.08	0.04
Coif4	0.04	0.09	0.06	0.05
Coif5	0.07	0.11	0.15	0.03

calculates the variance of X and P indicates the plaintexts that the adversaries can access. Since the target sensitive variables Z can usually be obtained by $Z = f(P, K)$ with f a bijective function and K representing the key, we can obtain that $Var(E(cA_i|Z)) = Var(E(cA_i|P))$, which proves the validity of Equation 5. For each wavelet parameters combination (functions Db7~Db10, Sim7~Sim10, Coif1~Coif5, with levels L7~L10), adjusted SNR will be calculated after WTP and WTA_{GMMs}.

$$SNR = \frac{Var(E(cA_i|P))}{Var(cA_i) - Var(E(cA_i|P))} \quad (5)$$

Table VII records the maximum SNR results for different wavelet parameter combinations. Since SNR is directly correlated with the attack ability, we select the ones whose SNR exceeds 0.1 as the candidate combinations for the final exhaustion. Therefore only 12 combinations remain for the final exhaustion, that is Db7~Db10 with L8~L9, Coif5 with L8~L9, Sym9 with L9, and Coif3 with L8.

E. Exhaustive Search

At this step, only 12 wavelet parameter combinations are selected for the final exhaustive search. Now we will elaborate on the method of exhaustive search to find the optimal wavelet parameters combination for our experimental set-up. We intend to utilize two metrics listed below to evaluate the attack effectiveness for each combination.

- MTD: *Minimum Traces to Disclose the correct key byte* [3], [42]. Since numerous traces tend to take much time and storage, it is always desirable to extract the secret key with less traces.
- MCV: *Maximum Correlation Value of the correct key byte* [3], [43]. This metric calculates the correlation value between the power consumption and the simulated model for key guess, which reflects the accuracy of attacks. The attack method is regarded to be better if its MCV is larger.

With each combination of wavelet parameters (total 12 combinations), a UWAF-based attack will be conducted on the traces. To evaluate the MTD of each combination, it is required to increase the amount of targeted traces one by one (randomly selected) and launch the same attack process repeatedly, until

TABLE VIII
MTD AND MCV OF THE CANDIDATE WAVELET
PARAMETERS COMBINATIONS

	L8		L9	
	MTD	MCV	MTD	MCV
Db7	467	0.2164	22	0.7589
Db8	532	0.1748	65	0.4091
Db9	453	0.2087	87	0.4033
Db10	377	0.2267	325	0.2163
Sym9	/	/	200	0.2606
Coif3	421	0.2167	/	/
Coif5	645	0.1729	122	0.4274

the attack succeeds in disclosing the key. It implies that each wavelet parameters combination requires multiple repeated attack processes. After searching exhaustively on the parameters combination candidates, the results of MTD and MCV are depicted in Table VIII. From Table VIII, it is demonstrated that the optimal combination is Db7 and L9, since its MTD (22 traces) is the least, while its MCV (0.7589) is the largest. We should highlight that this combination presents the largest SNR as well in Table VII, while with relatively large (but not the largest) RAD as in Table V. It implies that RAD is not a precise metric to indicate an optimal wavelet combination, but is able to perform an initial coarse-grained screening as in Section IV-C. Regarding SNR, an aggressive strategy is to determine the optimal combination exactly by SNR since it directly correlates to attack ability. However, we still recommend leveraging SNR as a fine-screening metric for fault tolerance. In the case where the SNR of all combinations are similar and relatively low (e.g., lower than 1), SNR will become insufficient to determine an optimal wavelet parameter combination.

The exhaustive search for optimal wavelet parameters is fulfilled at this step, however we intend to carry out another exhaustive search to check the screening strategy and verify the optimal parameters. Since for decomposition level, the coarse-grained screened L7~L10 are verified by practical attack experiments and the fine-grained screened L8~L9 are selected by SNR, meanwhile with the decomposition performance illustrated in Fig. 4 as an auxiliary proof, there is no need to check the selection strategy of decomposition level. According to MTD and MCV in Table VIII, L9 outperforms L8 significantly and thus L9 can be exactly selected as the optimal decomposition level. Based on it, we intend to combine all the wavelet basis functions belonging to DbN, SymN and CoifN families with level 9 to check if the coarse-grained screening strategy for the wavelet basis function is valid. Meanwhile, this can also verify if Db7 (combined with L9) can help to achieve the best attack performance.

The attack results are presented in Table IX below. It can be confirmed from Table IX that the combination of Db7 and level 9 is indeed the optimal combination in our case, with the least number of traces required and the largest correlation coefficient. The results also prove that the coarse-grained screening strategy for the wavelet basis function is valid, since it greatly reduces the computational cost without missing the optimal wavelet parameters. Here, we can conclude that the optimal wavelet

TABLE IX
EXHAUSTIVE SEARCHING
RESULTS AT
DECOMPOSITION L9

		MTD	MCV
Db1	L9	2021	0.0948
Db2	L9	104	0.3739
Db3	L9	58	0.5315
Db4	L9	552	0.1914
Db5	L9	1651	0.0945
Db6	L9	130	0.3558
Db7	L9	22	0.7589
Db8	L9	65	0.4091
Db9	L9	87	0.4033
Db10	L9	325	0.2163
Sym1	L9	2083	0.0825
Sym2	L9	102	0.3714
Sym3	L9	29	0.7094
Sym4	L9	128	0.3855
Sym5	L9	101	0.4043
Sym6	L9	41	0.5906
Sym7	L9	304	0.2169
Sym8	L9	532	0.1861
Sym9	L9	200	0.2606
Sym10	L9	124	0.3441
Coif1	L9	108	0.3782
Coif2	L9	262	0.2489
Coif3	L9	286	0.2426
Coif4	L9	440	0.2057
Coif5	L9	122	0.4274

parameters in our case are Db7 and L9, with the specific metrics that MTD=22, MCV=0.7589.

In conclusion, we leverage exhaustive search to find the optimal wavelet parameters thanks to the automation and unification of UWAF. Since the wavelet parameters are fully guided by experimental results, we believe the selected wavelet parameters (Db7 and L9) will be valid for similar attack scenarios. To mitigate computational overhead for exhaustion, we propose some approaches to reduce the scope of exhaustion to a reasonable range, which are verified by practical experiments as well. We believe that the above attack process could be regarded as a paradigm of exhaustive search for wavelet parameters leveraging UWAF.

V. EVALUATION AND COMPARISON

Guided by the experimental exhaustive search, Db7 and level 9 have been selected as wavelet parameters since they enable the best attack result for UWAF. And in this section, we conduct a comprehensive evaluation to compare UWAF against some counterpart framework-level methods. Since UWAF is carried out in the wavelet domain, time-domain and frequency-domain attack frameworks naturally become the opponents for comparison. As in [3], we select the *Standard Attack Procedure* (SAP) [1] and *Frequency Attack Procedure* (FAP) [26], which represents the time-domain and frequency-domain attack, respectively. Certainly to show the improvement of UWAF over WAF, the wavelet-domain WAF proposed in [3] will be compared as well.

To achieve a sound evaluation, in addition to MTD and MCV introduced in Section IV-E, we add another two metrics listed

TABLE X
ATTACK PERFORMANCE FOR DIFFERENT FRAMEWORKS

	SAP [1]	FAP [26]	WAF [3]	UWAF (this paper)
MTD	6865	993	463	22
MCV	0.073	0.181	0.299	0.759
TOP(sec)	22562.98	241.93	322.66	183.38
MTR(sec)	2809.2	6020.45	103.48	0.89

below to assess the attack performance, which focus on illustrating the computational cost.

- TOP: *Time Of Pre-processing of power traces* [3]. Before the direct attack for key recovery, the traces pre-processing period will take some non-negligible time. The attack with less processing time is always preferred.
- MTR: *Minimum Time to Recover a key byte correctly* [44]. The time for key extraction itself is also an important metric. An attack with less time for key extraction is commonly regarded as more efficient.

To calculate TOP, we should first define the pre-processing stage of each attack framework. SAP consists of three components in the time domain: *Standard Traces Denoising* (STD), *Standard Traces Alignment* (STA) and *Standard Key Extraction* (SKE) (for more details please refer to [1], [3]). Hence, its pre-processing stage includes both STD and STA components. While considering FAP, since alignment operation is not required in the frequency domain, STD is the only component in the pre-processing stage of FAP (for more details please refer to [3], [26]). Recall that both WAF and UWAF contain wavelet-based denoising components (WTD for WAF while WTP for UWAF) and wavelet-based alignment components WTA. Hence they both have two steps for trace pre-processing. It is worth noting that TOP records the time of the complete pre-processing stage for each attack framework.

The experimental setup has been introduced before in Section IV-A. And we should highlight that we choose such experimental set-up and evaluation metrics for a fair and direct comparison with WAF [3]. The attack results for all frameworks are listed in Table X, and the results of SAP, FAP and WAF are directly taken from those of Table VIII in [3]. Recall that the metrics MTD and MCV are utilized to illustrate the ability of key recovery, while TOP and MTR (in seconds) imply the computational cost of trace processing. It is demonstrated in Table X that UWAF outperforms extremely the other three attack frameworks from both attack ability and computational overhead perspectives. In terms of the attack ability, UWAF only requires 22 traces for successful key recovery, which is almost 20 times less than WAF and 312 (resp. 45) times less than SAP (resp. FAP). Other than reducing the required number of traces, its correlation value is up to 0.759, which is much higher than that of the other three frameworks. In terms of the computation overhead, it takes only 183.38 seconds for UWAF to process 9985 traces. In comparison, it has to take about 22562.98, 241.93 and 322.66 seconds for SAP, FAP and WAF, respectively. Compared with WAF, TOP for UWAF is almost halved. The reduced pre-processing time for UWAF is due to two factors: on one hand, the number of wavelet

decomposition halves and the signal re-construction process is accordingly erased; on the other hand, the automated clustering approach replaces the manually involved threshold method to align traces, helping to cut down the calculation cost. The time to disclose the key byte correctly (measured in MTR) is also reduced significantly. This is actually a bonus of MTD. Since much fewer traces are required for a successful attack in UWAF, it naturally needs less time for the attack process, resulting in smaller MTR.

In a short summary, UWAF performs the best among all the selected attack frameworks in terms of both attack effectiveness and computational cost. Especially compared with WAF, the improvement is significant. Based on the evaluations and comparisons, UWAF could be a better choice when breaking the countermeasure of random delay insertion.

VI. DISCUSSION AND CONCLUSION

In this paper, we focus on the attack against random delay insertion. It is worth noting that UWAF is also able to break Random Propagation Delay (RPD) and Random Clock Jitters (RCJ). Regarding RPD, it equals RDI-protected implementations with random delays inserted at the beginning. And for RCJ, the slight difference lies in the attack model utilized in WKE. That is, Hamming Weight model should be replaced by Hamming Distance model.

To defend against UWAF, the encryption designer can add other *hiding* countermeasures, such as *shuffling* [1]. Since the order of instructions for encryption is randomly executed by *shuffling*, no waveform part of traces should be eliminated in the alignment stage, which enables the WTA of UWAF to be ineffective. Moreover, *masking* [1] strategy can be combined with RDI as well to protect the encryption process from UWAF. Although WKE can be tuned by improving the direct CPA to higher-order CPA to attack masked implementations, the random shares of sensitive variables Z will render the SNR metric ineffective, leading to costly exhaustive search.

Since both UWAF and WAF [3] are framework-level attacks conducted in the wavelet domain, we should highlight their difference before the conclusion of this paper. There totally exists four different points between WAF and UWAF.

Difference in Denoising. Both WAF and UWAF include a denoising component. However, WTD of WAF is to eliminate the noise of cD , which will be utilized in the following signal reconstruction, while in UWAF, cD is discarded in WTP.

Difference in Alignment. WAF leverages the threshold method to differentiate the normal encryption part and random delays part, which requires a manual-driven set for the initial range of threshold. Such method actually hinders the automation of the attack program, increasing the process time. For automation, UWAF exploits the advanced machine-learning-based clustering method (GMMs clustering algorithm) to align the traces automatically and in an efficient fashion.

Unification of UWAF. Since the denoising and alignment stage of WAF require different wavelet parameter combinations, two wavelet decompositions are conducted in WAF and thus WTD and WTA exist in different wavelet domains. Recall that

the denoising stage of WAF is to reduce the noise in cD , while cD is directly discarded in UWAF. Hence the extra wavelet parameter combination for cD denoising is no longer required in UWAF, resulting in that only once wavelet decomposition for alignment is necessary. Moreover, since the alignment and the subsequent attack of both WAF and UWAF are performed on cA_i only, the removal of wavelet decomposition for denoising cD will not affect the attack ability of UWAF. Reducing once wavelet decomposition, another advantage of UWAF over WAF is that it reduces the extra re-construction (IDWT) from wavelet domain to time domain, which tends to introduce errors into cA_i . In a nutshell, UWAF unifies the wavelet-based analysis within the same wavelet domain without reducing the attack ability, halving the number of wavelet decompositions and accordingly decreasing the computational cost greatly.

Difference in Wavelet Parameters Selection. In WAF, both WTD and WTA require the selection of wavelet parameters. The wavelet parameters of WTD refer to the previous literature work, while the selection of WTA is based on theoretical derivation. However, the wavelet parameters determination of UWAF is completely guided by experimental results, which is more related to practice than WAF. And the experimental results in Section V also prove that UWAF outperforms WAF in practice.

At last, we give a summary of our work. Firstly, we devise a new attack framework UWAF which is unified and has an automatic execution flow. The improvement is significant in terms of both attack ability and computational overhead. In addition, we also recommend finding the optimal wavelet parameters by exhaustive search in the absence of a sound theoretical model and provide an example leveraging UWAF (which fits the exhaustion perfectly) on our physical observations. Meanwhile, we propose some approaches to reduce the range for exhaustive search which are verified by practical experiments to ease the computational complexity. By exhaustive search, we find a new combination of wavelet parameters which is never considered before but is optimal in our case. Finally, the evaluation and comparison with three typical framework-level attacks verify that UWAF could be a better choice when breaking implementations protected by random delay insertions.

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